



BLUETEAMERS
— DEFEND. DETECT. RESPOND. —

SPLUNK SIEM

LOG ANALYSIS PROJECTS

— A PRACTICAL NOTEBOOK FOR SOC ANALYSTS —



Collect. Analyze. Detect. Respond.
Turning Logs into Security Insights.

PREPARED BY
BLUETEAMERS


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blueteamers.io

SPLUNK ENTERPRISE

WHAT IS SPLUNK?

Splunk is a powerful SIEM and log analytics platform that collects, indexes, and analyzes machine data in real-time.

WHY SOC ANALYSTS USE SPLUNK?

- Collects logs from multiple sources
- Real-time monitoring and alerting
- Powerful search (SPL)
- Dashboards and visualizations
- Threat detection and investigation

SPLUNK ENTERPRISE vs SPLUNK CLOUD

SPLUNK ENTERPRISE	SPLUNK CLOUD
• Self-hosted • Full control • Requires infrastructure • Free for up to 500MB/day	• Cloud hosted by Splunk • Quick to start • Subscription based • Auto updates & scaling

SYSTEM REQUIREMENTS (MINIMUM)

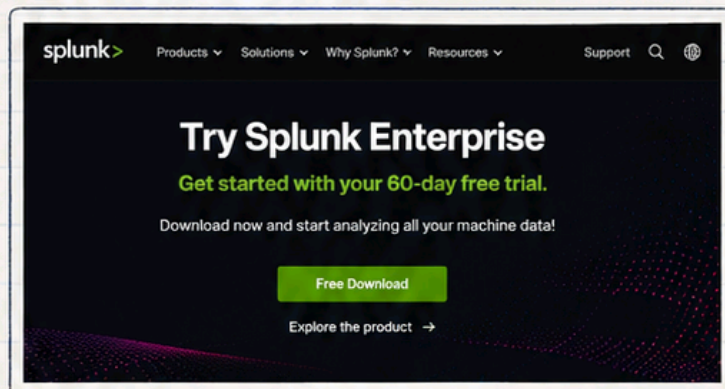
- CPU: 2 cores
- RAM: 8 GB
- Disk: 40 GB
- OS: Windows / Linux / macOS

DOWNLOAD SPLUNK ENTERPRISE

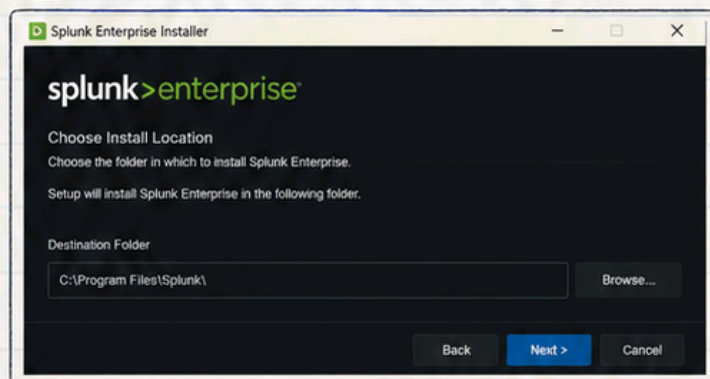
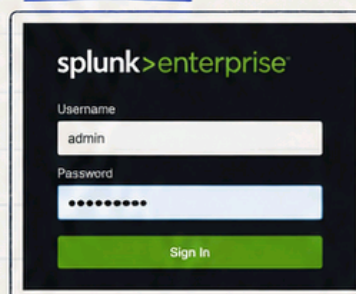
- ① Go to Splunk official website.
- ② Click on "Try Splunk Enterprise".
- ③ Choose "Free Download".
- ④ Select your OS and download.



TIP: Splunk Enterprise is free for up to 500MB of data per day.

SPLUNK DOWNLOAD PAGE

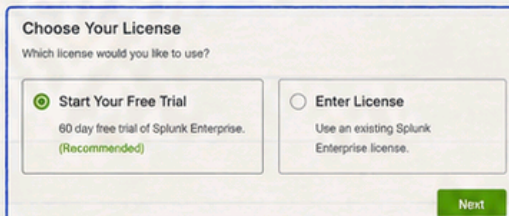
- ★ Select "Free Download" under Splunk Enterprise.

INSTALLATION WINDOW (WINDOWS)FIRST LOGINDEFAULT CREDENTIALS

- Username: admin
- Password: changeme

IMPORTANT

Change the default password after first login for security.

LICENSE SELECTION

- ★ Select "Start Your Free Trial" and click Next.





OVERVIEW

After installing Splunk, we need to configure basic settings, add data, and verify that logs are being indexed. Follow the steps below.

SPLUNK ARCHITECTURE (SIMPLE VIEW)



KEY CONCEPTS

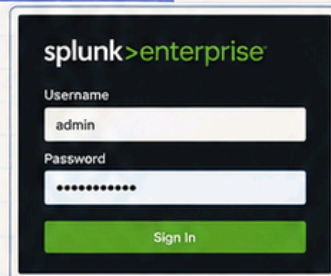
- Index : Where data is stored.
- Source : The origin of the log file.
- Sourcetype : Type/category of the data (e.g., dns, http).
- Host : The system that generated the logs.
- Event : A single log entry.

STEP 1: ACCESS SPLUNK WEB INTERFACE

1. Open your browser.
2. Enter the URL:

`http://localhost:8000`
(or your server IP)

3. You will see the Splunk login page.



STEP 2: FIRST LOGIN

- ① Default Username: admin
- ② Default Password: changeme
- ③ Click "Sign In".



IMPORTANT: Change the default password after first login!



STEP 3: CHANGE ADMIN PASSWORD

1. Go to Settings > Users and Authentication > Change Password.
2. Enter your current password (changeme).
3. Enter a new strong password and confirm.
4. Click "Change Password".



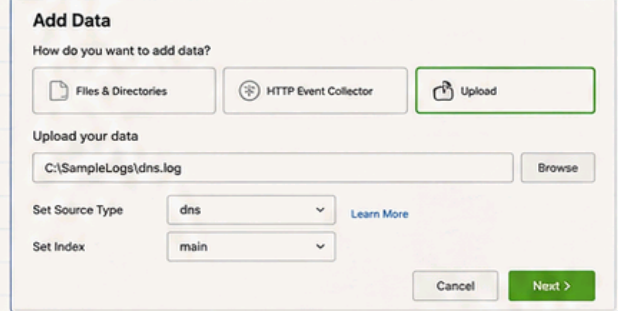
STEP 7: CREATE A SIMPLE DASHBOARD

1. Go to Dashboards > Create New Dashboard.
2. Click "Add Panel" > "Search".
3. Enter a search (e.g., `index=main | timechart count`).
4. Click "Save" and name your dashboard.



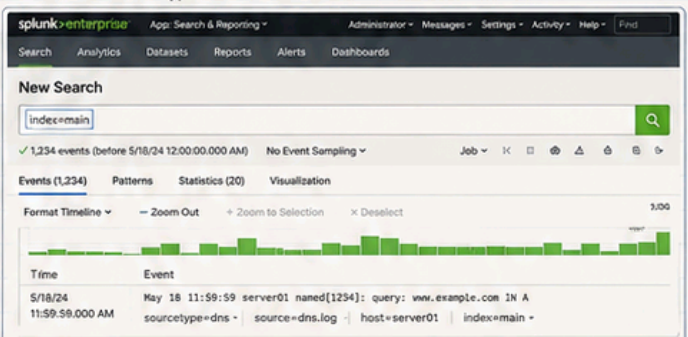
STEP 4: ADD DATA (UPLOAD SAMPLE LOGS)

1. Click on Settings (gear icon) > Add Data.
2. Select "Upload".
3. Drag and drop your sample log file or click "Select files" to browse.
4. Set the following:
 - Index : main (or create new index)
 - Source type : Select or enter the correct sourcetype (e.g., dns, http, ftp)
5. Click "Next".
6. Review the settings and click "Submit".



STEP 5: VERIFY DATA (SEARCH & REPORTING APP)

1. Go to Search & Reporting app.
2. In the search bar, type: `index=main`
3. Click the green search button.
4. Events should appear on the screen.



STEP 6: EXPLORE YOUR DATA

Try some basic SPL searches:

QUERY	PURPOSE	EXAMPLE RESULT
<code>index=main head 10</code>	Show first 10 events	
<code>index=main stats count by sourcetype</code>	Count events by sourcetype	
<code>index=main stats count by host</code>	Count events by host	
<code>index=main timechart count</code>	Events over time	

COMMON MISTAKES

- Wrong sourcetype selected
- Logs not added to the correct index
- Time range set incorrectly
- Firewall blocking port 8000



DNS LOG ANALYSIS



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This project provides a step-by-step guide for analyzing DNS (Domain Name System) log files using Splunk SIEM.



WHAT ARE DNS LOGS?

DNS (Domain Name System) logs record queries and responses between clients and DNS servers. They show which domains were requested, the response returned, and other metadata like timestamps, client IPs, and query types.

WHY DNS LOGS MATTER IN SOC?

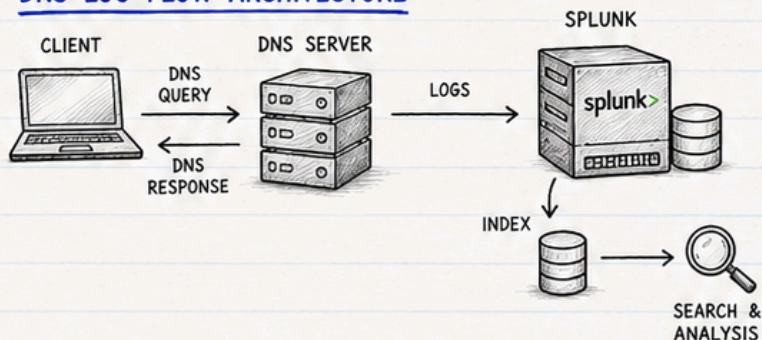
- Identify suspicious or malicious domain lookups
- Detect DNS tunneling and data exfiltration
- Discover compromised hosts communicating with C2 domains
- Investigate phishing, malware, and beaconing activity
- Baseline normal DNS behavior in your environment

PREREQUISITES

- ☒ Splunk instance is installed and configured
- ☒ DNS log data sources are configured to forward logs to Splunk
- ☒ DNS server is generating logs (e.g., BIND, Windows DNS, Infoblox, Cisco Umbrella)
- ☒ Universal Forwarder installed (if logs are on a separate machine)
- ☒ Proper sourcetype configured for DNS logs (e.g., dns, bind, windows:dns)



DNS LOG FLOW ARCHITECTURE



COMMON DNS LOG SOURCES

- BIND DNS Server
- Windows DNS Server
- Cisco Umbrella / OpenDNS
- Infoblox DNS
- Zeek (dns.log)
- Palo Alto Networks Firewall (DNS Security logs)

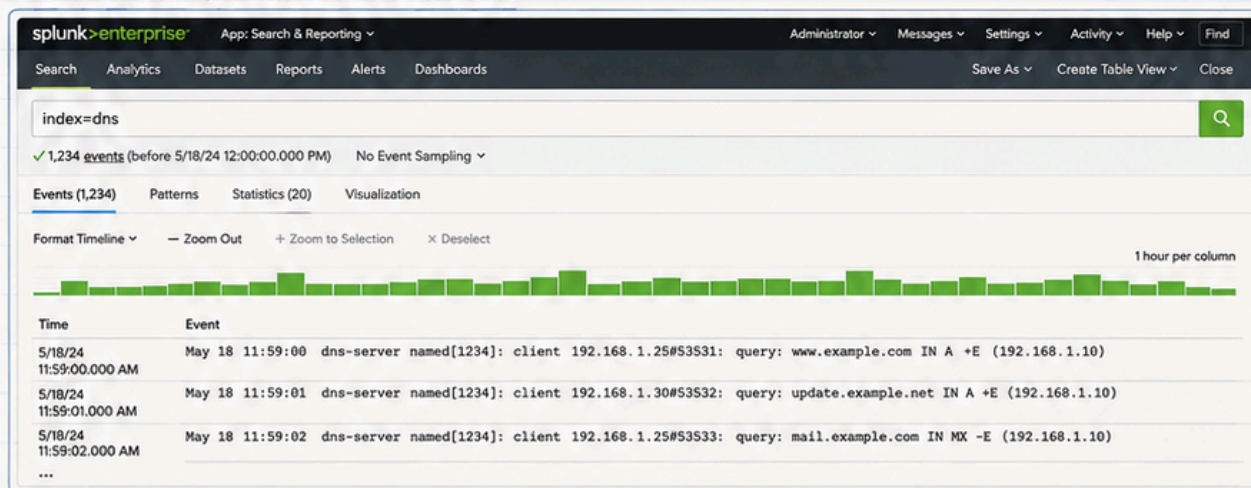


SAMPLE DNS LOG (BIND FORMAT)

May 18 11:59:00 dns-server named[1234]: client 192.168.1.25#53531:
query: www.example.com IN A +E (192.168.1.10)

Timestamp Client IP (Source) Query Name (Domain) Query Type (A, AAAA, MX) DNS Server IP (Responder)

VIEW IN SPLUNK (SEARCH & REPORTING APP)



KEY TAKEAWAYS

DNS logs are one of the most valuable data sources for threat detection. Unusual DNS queries can be an early indicator of compromise or malicious activity.





1. UPLOADING DNS LOGS TO SPLUNK

- Go to **Settings** (gear icon) > **Add Data**.
- Click **"Upload"**.
- Drag and drop your DNS log file (e.g., dns.log) or click **"Select files"**.
- Set the following:
 - Index : dns (create new or use existing)
 - Source type : dns
 - Host : dns-server
- Click **"Next"** and review the settings.
- Click **"Submit"**.

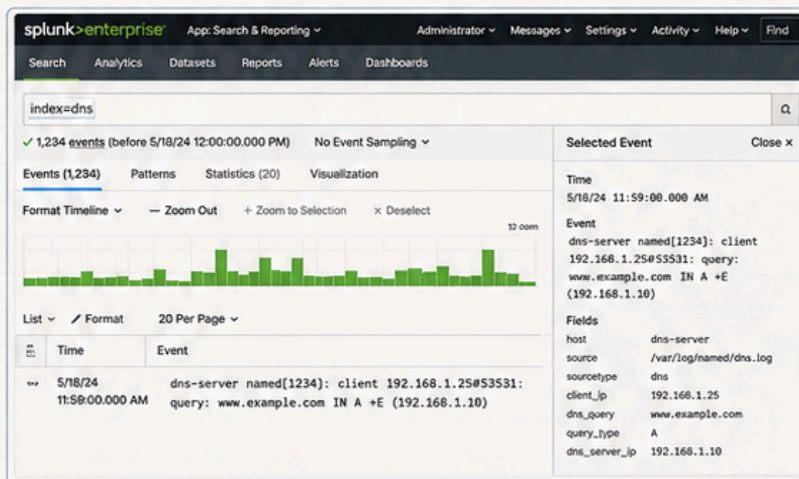
SAMPLE DNS LOG (BIND FORMAT)

May 18 11:59:00 dns-server named[1234]: client 192.168.1.25#53531: query: www.example.com IN A +E (192.168.1.10)

↑ ↑ ↑ ↑ ↑
 Timestamp Client IP Query Name Query Type DNS Server IP
 (Source) (Domain) (A, AAAA, MX) (Responder)

2. VERIFY DATA IN SPLUNK

- Go to **Search & Reporting** app.
- In the search bar, type:
- You should see DNS events in the results.
- Click on an event to view all extracted fields.



3. USEFUL SPL QUERIES FOR DNS LOG ANALYSIS

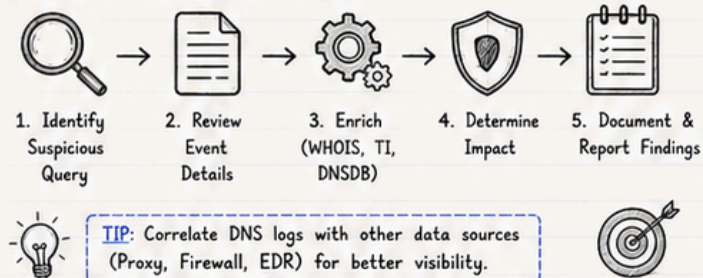
#	QUERY	PURPOSE	EXAMPLE RESULT (DESCRIPTION)
1	index=dns	Show all DNS logs	Displays all DNS events ingested.
2	index=dns stats count by query	Top queried domains	Shows most queried domains in the network.
3	index=dns stats count by src_ip	Top DNS clients	Shows clients generating most DNS queries.
4	index=dns stats count by query_type	Query types distribution	Shows A, AAAA, MX, TXT, NS query counts.
5	index=dns where query_type="A" stats count by query	Top A record lookups	Top IPv4 lookups.
6	index=dns where isnull(query)	Find failed/empty queries	Detect unusual or malformed queries.
7	index=dns stats count by rcodename	DNS response codes	Shows NOERROR, NXDOMAIN, SERVFAIL, etc.
8	index=dns rcodename=NXDOMAIN stats count by query	Detect potential DNS enumeration	Lists domains not found (possible recon).
9	index=dns timechart count by query_type	DNS queries over time	Visualizes DNS traffic trends.
10	index=dns top limit=10 query	Top 10 queried domains	Quick list of top 10 domains.

4. COMMON FINDINGS

- High volume of queries to a single domain (possible tunneling)
- Many NXDOMAIN responses (possible enumeration)
- Queries to suspicious domains (malware, phishing)
- Unusual query types (TXT, NULL, ANY)
- DNS queries outside business hours



5. INVESTIGATION WORKFLOW



FTP LOG ANALYSIS



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This project guides you through analyzing FTP (File Transfer Protocol) log files using Splunk SIEM.



WHAT ARE FTP LOGS?

FTP logs record file transfer activities between clients and servers. They include user logins, commands, file uploads/downloads, successes, failures, and session details.

PREREQUISITES

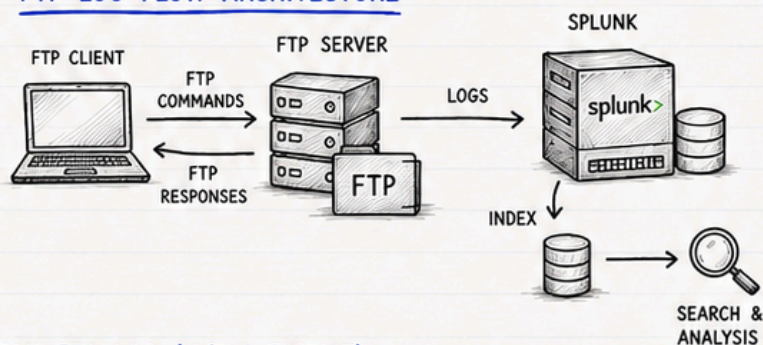
- ☒ Splunk instance is installed and configured
- ☒ FTP log data sources are configured to forward logs to Splunk
- ☒ FTP server is generating logs (e.g., vsftpd, ProFTPD, Windows IIS FTP)
- ☒ Universal Forwarder installed (if logs are on a separate machine)
- ☒ Proper sourcetype configured for FTP logs (e.g., ftp, vsftpd, proftpd)



WHY FTP LOGS MATTER IN SOC?

- Detect brute force login attempts
- Identify unauthorized file transfers
- Monitor suspicious downloads or uploads
- Detect data exfiltration
- Ensure compliance and track user activities

FTP LOG FLOW ARCHITECTURE



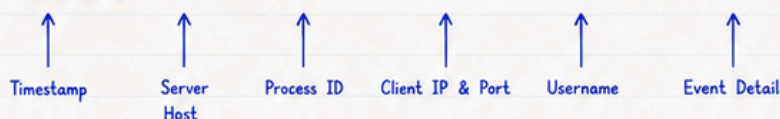
COMMON FTP LOG SOURCES

- vsftpd (Linux)
- ProFTPD (Linux)
- IIS FTP (Windows)
- FileZilla Server
- Pure-FTPd

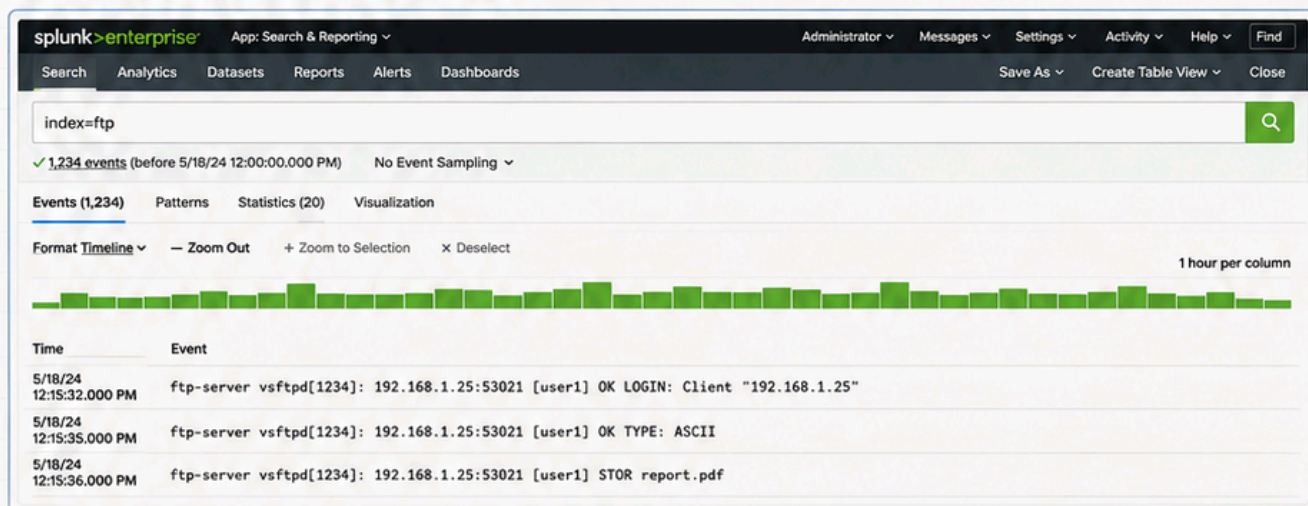


SAMPLE FTP LOG (vsftpd FORMAT)

May 18 12:15:32 ftp-server vsftpd[1234]: 192.168.1.25:53021 [user1] OK LOGIN: Client "192.168.1.25"



VIEW IN SPLUNK (SEARCH & REPORTING APP)



KEY TAKEAWAYS

FTP logs help SOC Analysts detect unauthorized access, brute force attempts, and suspicious file transfers that may lead to data breaches.



1. UPLOAD FTP LOGS TO SPLUNK

1. Go to **Settings** (gear icon) > **Add Data**.
2. Click **"Upload"**.
3. Drag and drop your FTP log file or click **"Select files"**.
4. Set the following:
 - Index : ftp (or create new index)
 - Source type : ftp (or custom ftp)
 - Host : ftp-server
5. Click **"Next"** and review the settings.
6. Click **"Submit"**.



Add Data

How do you want to add data?

☐ Files & Directories
 ☐ HTTP Event Collector
 ☒ Upload

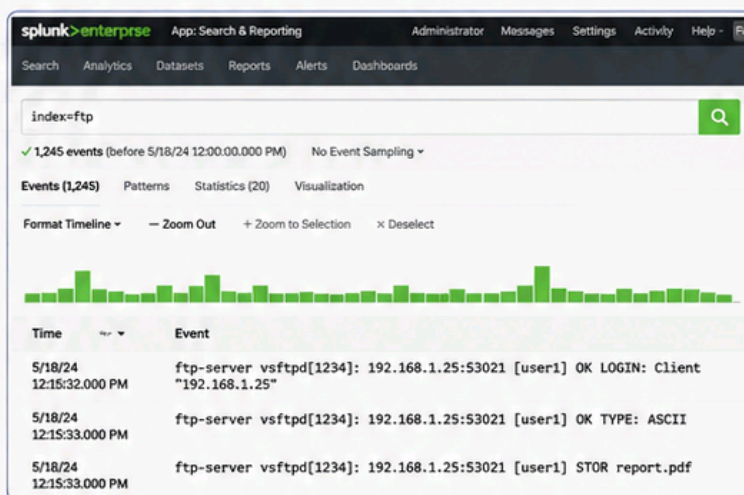
Upload your data

Set Source Type

Set Index

2. VERIFY LOGS IN SPLUNK

1. Go to **Search & Reporting** app.
2. In the search bar, type:
3. You should see FTP events in the results.



3. USEFUL SPL QUERIES FOR FTP LOG ANALYSIS

#	QUERY	PURPOSE
1	index=ftp	Show all FTP events
2	index=ftp stats count by user	Top users
3	index=ftp stats count by src_ip	Top client IPs
4	index=ftp action=login	Successful logins
5	index=ftp action=failed	Failed login attempts
6	index=ftp top filename	Most transferred files
7	index=ftp where action="upload"	Show upload activities
8	index=ftp where action="download"	Show download activities
9	index=ftp timechart count	FTP activity over time
10	index=ftp stats dc(session_id) by user	Unique sessions by user

4. COMMON ATTACK SCENARIOS



Brute Force Login

Attackers try multiple passwords to gain access.



Anonymous FTP Login

Anonymous access can lead to unauthorized downloads.



Unauthorized File Upload

Attackers upload malware or web shells.



Data Exfiltration

Large file downloads may indicate data theft.



Malware Upload

Malicious files uploaded to the server.

5. INVESTIGATION WORKFLOW



Alert Triggered



Review
Login Attempts



Identify
Source IP



Review Uploaded /
Downloaded Files



Correlate with
Firewall Logs



Contain User &
Block IP



Create Incident
and Report

6. SOC ANALYST NOTES

Look for the following suspicious indicators:

- | | |
|--|---|
| ☒ Multiple failed login attempts | ☒ Large file downloads |
| ☒ Logins outside business hours | ☒ Large file uploads |
| ☒ Anonymous FTP access | ☒ Unusual file extensions (.exe, .sh, .php) |
| ☒ Unexpected user accounts | ☒ Connections from unknown IPs/countries |



7. KEY TAKEAWAY

FTP logs provide valuable visibility into file transfer activities. Proper monitoring helps detect brute force attacks, unauthorized access, malicious uploads, and potential data exfiltration.



HTTP LOG ANALYSIS



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This project helps you analyze HTTP (Hypertext Transfer Protocol) log files to detect suspicious web activity and attacks.

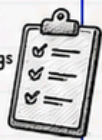


1. WHAT ARE HTTP LOGS?

HTTP logs are generated by web servers like Apache, Nginx, and IIS. They record details about client requests, responses, status codes, URLs accessed, user agents, and more.

PREREQUISITES

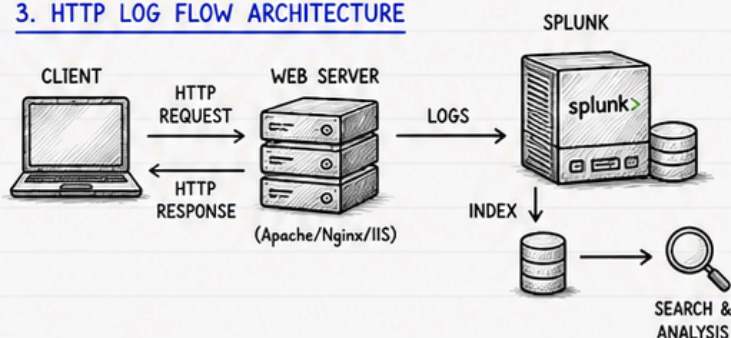
- ☒ Splunk instance is installed and configured
- ☒ HTTP log sources are configured to forward logs to Splunk
- ☒ Web server (Apache/Nginx/IIS) is generating logs
- ☒ Universal Forwarder installed (if logs are on a separate machine)
- ☒ Proper sourcetype configured for HTTP logs (e.g., apache, nginx, iis)



2. WHY HTTP LOGS MATTER IN SOC?

- Identify web attacks (SQLi, XSS, RCE, LFI, etc.)
- Detect reconnaissance and directory brute-forcing
- Monitor web shell uploads
- Identify suspicious user agents and bots
- Track errors and performance issues
- Correlate with other logs for threat detection

3. HTTP LOG FLOW ARCHITECTURE



4. COMMON HTTP LOG SOURCES

- Apache Access Logs
- Nginx Access Logs
- IIS Logs
- Web Application Logs
- Reverse Proxy Logs



5. SAMPLE HTTP LOG (APACHE COMBINED FORMAT)

```
192.168.1.25 - - [18/May/2024:12:45:32 +0530] "GET /login.php HTTP/1.1" 200 512
"https://example.com/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36
(KHTML, like Gecko) Chrome/124.0.0.0 Safari/537.36"
```

↑ Client IP ↑ Idendt ↑ User ↑ Timestamp ↑ Request Method & URL ↑ HTTP Version ↑ Status Code ↑ Bytes Sent ↑ Referrer URL ↑ User Agent

6. VIEW IN SPLUNK (SEARCH & REPORTING APP)

splunk>enterprises App: Search & Reporting

Search Analytics Datasets Reports Alerts Dashboards

index>http

✓ 2,345 events (before 5/18/24 12:00:00.000 PM) No Event Sampling

Events (2,345) Patterns Statistics (20) Visualization

Format Timeline Zoom Out Zoom to Selection Deselect

1 hour per column

Time	Event
5/18/24 12:45:32.000 PM	192.168.1.25 - - [18/May/2024:12:45:32 +0530] "GET /login.php HTTP/1.1" 200 512 "https://example.com/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/124.0.0.0 Safari/537.36"
5/18/24 12:45:33.000 PM	192.168.1.30 - - [18/May/2024:12:45:33 +0530] "POST /submit HTTP/1.1" 302 620 "-" "curl/7.81.0"
5/18/24 12:45:34.000 PM	192.168.1.22 - - [18/May/2024:12:45:34 +0530] "GET /admin/ HTTP/1.1" 403 489 "-" "Mozilla/5.0 (X11; Linux x86_64)"

7. KEY TAKEAWAYS



HTTP logs are critical for detecting web-based attacks, unauthorized access, enumeration, and suspicious behavior. Always monitor status codes, URLs, and user agents to identify anomalies early.



8. COMMON HTTP STATUS CODES

- 200 - OK
- 301/302 - Redirect
- 400 - Bad Request
- 401 - Unauthorized
- 403 - Forbidden
- 404 - Not Found
- 500 - Internal Server Error
- 503 - Service Unavailable



HTTP LOG ANALYSIS (HANDS-ON)



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Let's perform hands-on analysis of HTTP logs to uncover suspicious web activity and attacks.



1. UPLOAD HTTP LOGS TO SPLUNK

- 1 Go to **Settings** (gear icon) > **Add Data**
- 2 Click **"Upload"**
- 3 Drag and drop your HTTP log file or click **"Select files"**.
- 4 Set Source Type: **http**
- 5 Set Index: **http**
- 6 Click **"Next"** and review the settings
- 7 Click **"Submit"**



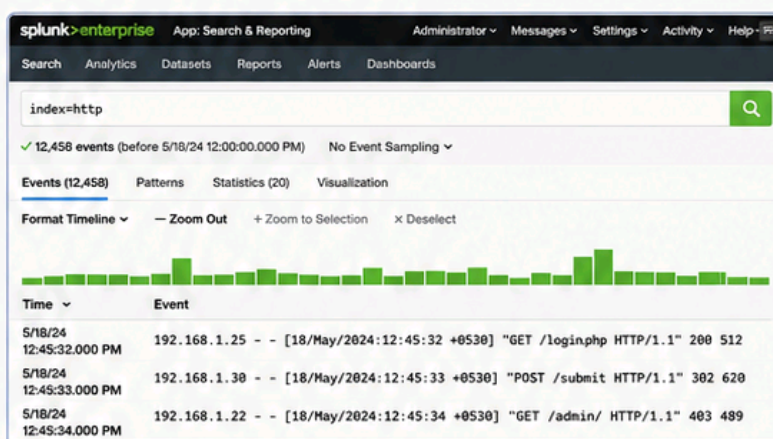
TIP: Use sourcetype **http** (or custom) for better parsing and search.

2. VERIFY LOGS IN SPLUNK

1. Go to **Search & Reporting** app.
2. In the search bar, type:

index=http

3. You should see HTTP events in the results.



3. USEFUL SPL QUERIES FOR HTTP LOG ANALYSIS

#	SPL QUERY	PURPOSE	EXAMPLE OUTPUT
1	index=http	Show all HTTP events	All HTTP traffic
2	index=http stats count by uri	Top requested URIs	/login.php, /index.html...
3	index=http stats count by status	Status code distribution	200, 404, 403, 500...
4	index=http status=404	Find all 404 errors	Not Found requests
5	index=http status=403	Find forbidden access	Access denied attempts
6	index=http method=POST	Find POST requests	Forms, logins, uploads
7	index=http "../"	Potential directory traversal	../, ..%2f, ..%5c
8	index=http top user_agent	Top user agents	Chrome, curl, bots...
9	index=http timechart count	Traffic over time	Web activity trend
10	index=http stats count by src_ip	Top source IPs	Most active IPs

4. COMMON HTTP ATTACKS TO DETECT



SQL Injection (SQLi)

Look for patterns like ' or 1=1 --, union select in URLs or POST data.



Cross Site Scripting (XSS)

Look for <script>, alert(1), javascript: in parameters.



Web Shell Upload

Look for uploads of .php, .asp, .jsp files.



Directory Traversal

Look for ../, ...%2f, ..%5c in URLs.



Brute Force / Login Attempts

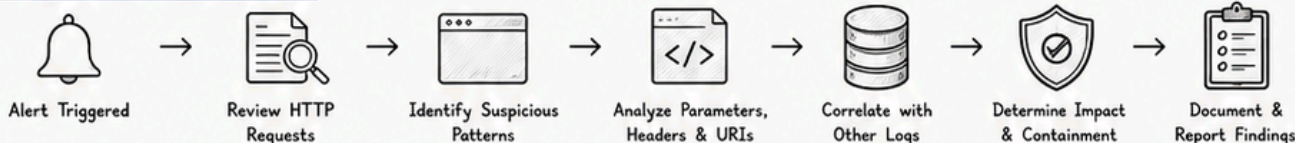
Multiple failed login attempts on login pages.



Reconnaissance

Automated scans, admin panel access, robots.txt, .env, backup files.

5. INVESTIGATION WORKFLOW



6. SOC ANALYST NOTES (WHAT TO LOOK FOR)

- ✓ Unusual URLs or parameters
- ✓ High number of 404 or 403 errors
- ✓ Suspicious file uploads
- ✓ SQLi or XSS patterns
- ✓ Directory traversal attempts
- ✓ Brute force on login pages
- ✓ Traffic from uncommon countries
- ✓ Abnormal spikes in requests



7. KEY TAKEAWAY

HTTP logs are essential for detecting web attacks such as SQLi, XSS, file uploads, and scanning activity. Monitoring URLs, status codes, methods, and user agents helps identify threats early.



SSH LOG ANALYSIS



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This project helps you analyze SSH (Secure Shell) logs to detect unauthorized access, brute force attacks, and suspicious activities on Linux/Unix systems.



1. WHAT ARE SSH LOGS?

SSH logs record authentication attempts, successful logins, failed logins, session activities, and user commands executed on servers.

PREREQUISITES

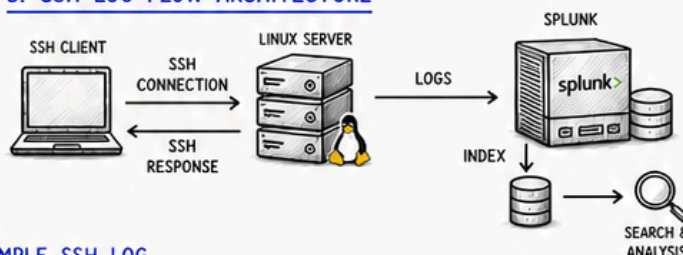
- ☒ Splunk instance is installed and configured
- ☒ SSH logs are forwarded to Splunk (via rsyslog/syslog or file input)
- ☒ Linux/Unix server generating logs
- ☒ Proper sourcetype configured for SSH logs (e.g., ssh, linux_secure, syslog)



2. WHY SSH LOGS MATTER IN SOC?

- Detect brute force login attempts
- Identify unauthorized access to critical servers
- Monitor privileged (root) logins
- Track use of SSH keys and unusual commands
- Detect lateral movement and persistence
- Ensure compliance and audit user activity

3. SSH LOG FLOW ARCHITECTURE



4. COMMON SSH LOG SOURCES

- /var/log/auth.log (Debian/Ubuntu)
- /var/log/secure (RHEL/CentOS)
- /var/log/sshhd.log
- Syslog / Rsyslog Forwarder
- Cloud Server Logs (AWS, Azure, GCP)



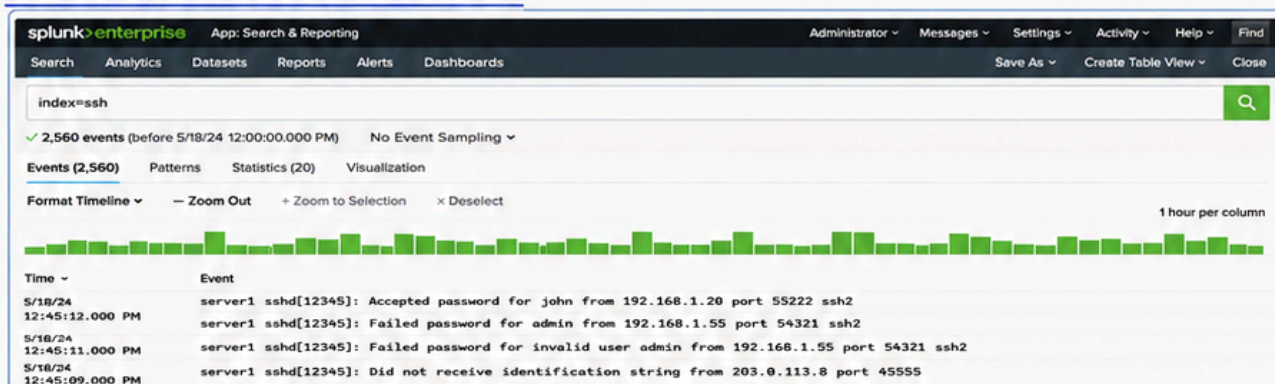
5. SAMPLE SSH LOG

```

May 18 12:45:10 server1 sshd[12345]: Failed password for invalid user admin
from 192.168.1.55 port 54321 ssh2
May 18 12:45:11 server1 sshd[12345]: Failed password for admin from 192.168.1.55
port 54321 ssh2
May 18 12:45:12 server1 sshd[12345]: Accepted password for john from 192.168.1.20
port 55222 ssh2
May 18 12:45:13 server1 sshd[12345]: session opened for user john by (uid=0)
  
```

Timestamp Host Process Message Username Source IP Port Method / Result

6. VIEW IN SPLUNK (SEARCH & REPORTING APP)



7. USEFUL SPL QUERIES FOR SSH LOG ANALYSIS

#	SPL QUERY	PURPOSE	EXAMPLE OUTPUT
1	index=ssh	Show all SSH events	All SSH activities
2	index=ssh action=failed	Failed login attempts	Failed password events
3	index=ssh action=accepted	Successful logins	Accepted password events
4	index=ssh stats count by src_ip	Top source IPs	Most active IPs
5	index=ssh stats count by user	Top users	Most active users
6	index=ssh top user	Most logins by user	Top login user
7	index=ssh timechart count	SSH activity over time	Traffic trend
8	index=ssh user=root	Root login attempts	Root access events

8. COMMON SSH ATTACK SCENARIOS



Brute Force Login
Attackers try multiple passwords to gain access.



Failed Login Attempts
Multiple failed attempts indicate brute force.



Root Login
Direct root login may indicate compromise.



SSH Key Abuse
Unauthorized use of valid SSH keys.



Suspicious Commands
Attackers execute malicious or unusual commands.

9. INVESTIGATION WORKFLOW



10. KEY TAKEAWAY

SSH logs are critical for detecting brute force attacks, unauthorized access, root logins, and suspicious activities on servers. Always monitor and respond quickly.



SSH LOG ANALYSIS (HANDS-ON)



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Let's perform hands-on analysis of SSH logs to detect brute force attacks, privilege abuse, and suspicious activities on Linux/Unix systems.



1. UPLOAD SSH LOGS TO SPLUNK

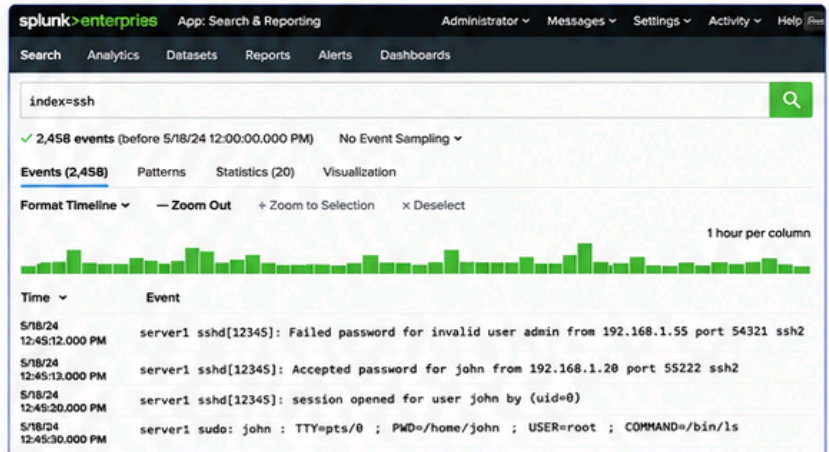
1. Go to Settings (gear icon) > Add Data
2. Click "Upload"
3. Drag and drop your SSH log file or click "Select files".
4. Set Source Type: `ssh`
5. Set Index: `ssh`
6. Click "Next" and review the settings
7. Click "Submit"



TIP: Use sourcetype `ssh` (or custom) for better parsing and search.

2. VERIFY LOGS IN SPLUNK

1. Go to Search & Reporting app.
2. In the search bar, type: `index=ssh`
3. You should see SSH events in the results.



3. USEFUL SPL QUERIES FOR SSH LOG ANALYSIS

#	SPL QUERY	PURPOSE	EXAMPLE OUTPUT
1	<code>index=ssh</code>	Show all SSH events	All SSH activities
2	<code>index=ssh action=failure</code>	Failed login attempts	Failed password events
3	<code>index=ssh action=success</code>	Successful logins	Accepted password events
4	<code>index=ssh stats count by src_ip</code>	Top source IPs	Most active IPs
5	<code>index=ssh stats count by user</code>	Top users	Most active users
6	<code>index=ssh top user</code>	Most logins by user	Top login users
7	<code>index=ssh timechart count</code>	SSH activity over time	Traffic trend
8	<code>index=ssh user=root</code>	Root login attempts	Root access events
9	<code>index=ssh where action="failure" stats values(uwer) by src_ip</code>	Users targeted by IP	Attacker targeting info
10	<code>index=ssh stats count by dest_port</code>	Top destination ports	Port usage

4. COMMON SSH ATTACK SCENARIOS



Brute Force Login

Attackers try multiple passwords to gain access.



Failed Login Attempts

Multiple failed attempts indicate brute force attacks.



Root Login

Direct root login may indicate compromise.



SSH Key Abuse

Unauthorized use or upload of SSH keys.



Privilege Escalation

Use of `sudo` or `su` to gain higher privileges.



Lateral Movement

SSH access to other internal systems.

5. INVESTIGATION WORKFLOW



6. SOC ANALYST NOTES (WHAT TO LOOK FOR)

- Multiple failed login attempts
- Logins outside business hours
- Root login from non-admin IPs
- Use of weak or default accounts
- Suspicious commands execution
- SSH access from unusual countries
- New or unknown SSH keys
- Privilege escalation using `sudo`/`su`
- High frequency of SSH connections
- Lateral movement to other servers

7. KEY TAKEAWAY

SSH logs are critical for detecting brute force attacks, unauthorized access, root logins, and privilege abuse. Continuous monitoring helps prevent compromise and secure your Linux/Unix systems.



SMTP LOG ANALYSIS (HANDS-ON)



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LET'S ANALYZE SMTP LOGS IN SPLUNK

1 UPLOAD SMTP LOGS TO SPLUNK

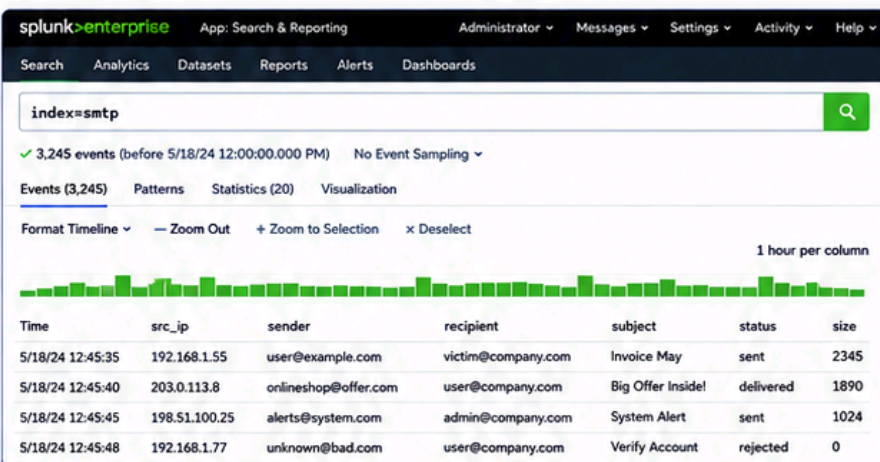
- Go to **Settings** (gear icon) > **Add Data**
- Click **"Upload"**
- Drag and drop your SMTP log file or click **"Select files"**
- Set **Source Type: smtp**
- Set **Index: smtp**
- Click **"Next"** and review settings
- Click **"Submit"**

Upload
SMTP Log File**TIP:** Use sourcetype `smtp` (or `custom`) for better parsing and search.

2 VERIFY SMTP LOGS

Search & Reporting app > Enter the search:

index=smtp



3 10 USEFUL SPL QUERIES FOR SMTP LOGS

#	SPL QUERY	PURPOSE
1	index=smtp	Show all SMTP events
2	index=smtp stats count by status	Count events by status (sent/delivered/failed)
3	index=smtp stats count by sender	Top senders
4	index=smtp stats count by recipient	Top recipients
5	index=smtp stats sum(size) by sender	Top data senders (by size)
6	index=smtp status="failed"	Failed email deliveries
7	index=smtp subject="*spam*"	Possible spam emails
8	index=smtp size>1000000	Large email detection
9	index=smtp timechart count by status	Email trends over time
10	index=smtp src_ip!<internal_ip>	External SMTP connections

4 DETECT PHISHING EMAILS



Look for suspicious subjects, links, and external senders.

```
index=smtp subject="*urgent*" OR subject="*verify*"
OR subject="*password*"
| stats count by sender, subject, src_ip
```

5 DETECT SPAM EMAILS



Identify potential spam based on keywords and patterns.

```
index=smtp subject="*free*" OR subject="*win*" OR subject="*offer*"
| stats count by sender, subject
```

6 DETECT MALWARE ATTACHMENTS



Detect emails with dangerous file types.

```
index=smtp (attachment_name="*.exe" OR "*.scr" OR "*.bat"
OR "*.js" OR "*.vbs")
| table _time, sender, recipient, subject, attachment_name
```

7 HEADER ANALYSIS

Analyze email headers for suspicious origins and spoofing.



```
index=smtp
| table _time, sender, subject, headers
| search headers="Received:"
```

8 SPF CHECK

Check SPF failures.

```
index=smtp spf="fail"
| stats count by sender
| sort - count
```

9 DKIM CHECK

Check DKIM failures.

```
index=smtp dkim="fail"
| stats count by sender
| sort - count
```

10 DMARC CHECK

Check DMARC failures.

```
index=smtp dmarc="fail"
| stats count by sender
| sort - count
```

11 INVESTIGATION WORKFLOW



12 SOC ANALYST NOTES



- Always validate sender and domain.
- Check for mismatched SPF, DKIM, DMARC.
- Analyze attachments in sandbox.
- Correlate with proxy, firewall & user activity.
- Report and block malicious senders.



13 KEY TAKEAWAY

SMTP log analysis helps SOC analysts detect email-based threats, prevent data loss, and ensure secure and reliable email communication in the organization.



DHCP LOG ANALYSIS


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UNDERSTANDING DHCP LOGS

DHCP logs help SOC analysts monitor IP address assignments, detect rogue DHCP servers, and identify suspicious network activity.



1 WHAT ARE DHCP LOGS?

DHCP (Dynamic Host Configuration Protocol) logs record events between DHCP clients and servers when IP addresses are assigned, renewed, or released.

2 WHY DHCP LOGS MATTER

- ✓ Track IP address allocation and usage
- ✓ Detect rogue DHCP servers
- ✓ Identify unauthorized devices
- ✓ Troubleshoot IP-related issues
- ✓ Support incident response & forensics



Key Takeaway

DHCP log analysis helps SOC analysts maintain network visibility and protect against unauthorized devices and IP conflicts.

3 PREREQUISITES



Splunk Enterprise installed & configured



DHCP Server (Windows / ISC DHCP / Cisco)



DHCP logs forwarded to Splunk



Sourcetype configured for DHCP logs

4 DHCP DORA PROCESS



DISCOVER

Client broadcasts DHCPDISCOVER to find servers.



OFFER

Server responds with DHCPOFFER (offered IP address).



REQUEST

Client requests the offered IP using DHCPREQUEST.



ACKNOWLEDGE

Server acknowledges with DHCPACK. Lease is active!

5 DHCP PORTS

PORT	PROTOCOL	DESCRIPTION
67	UDP	DHCP Server
68	UDP	DHCP Client

6 SAMPLE DHCP LOG (WINDOWS DHCP SERVER)

05/18/2024 12:45:32 PM DHCPSEVER Microsoft-Windows-DHCP-Server 1042
The DHCP server offered IP address 192.168.1.105 to client with MAC address 00-50-56-9A-1B-2C.

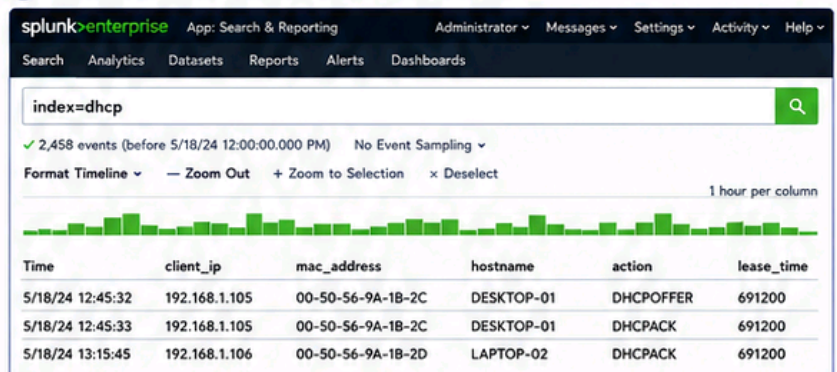
05/18/2024 12:45:33 PM DHCPSEVER Microsoft-Windows-DHCP-Server 1044
The DHCP server assigned IP address 192.168.1.105 to client with MAC address 00-50-56-9A-1B-2C. Lease duration 8 days.

7 COMMON DHCP LOG SOURCES

- Windows DHCP Server Logs (Event ID 1040–1058)
- ISC DHCP Server Logs
- Cisco DHCP Snooping Logs
- Fortinet / Palo Alto DHCP Logs
- Linux / Syslog DHCP Logs



8 VIEW IN SPLUNK (SEARCH & REPORTING APP)



9 IMPORTANT DHCP LOG FIELDS

FIELD	DESCRIPTION
timestamp	Time of the event
client_ip	IP address assigned to the client
mac_address	MAC address of the client
hostname	Hostname of the client (if available)
action	DHCP action (Discover, Offer, Request, Ack, NAK, Release)
server_ip	DHCP server IP address
lease_time	Lease duration (in seconds)
message_type	DHCP message type (DHCPOFFER, DHCPACK, etc.)

10 KEY TAKEAWAY



DHCP log analysis helps SOC analysts maintain network visibility, detect rogue DHCP servers, identify unauthorized devices, and ensure proper IP address management across the network.



FIREWALL LOG ANALYSIS


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UNDERSTANDING FIREWALL LOGS

Firewall logs capture network traffic allowed, blocked, or monitored by the firewall based on security policies and rules.



1 WHAT ARE FIREWALL LOGS?

Firewall logs record events such as allowed or blocked connections, rule hits, policy violations, dropped packets, and security threats detected by the firewall.

2 WHY FIREWALL LOGS MATTER

- ✓ Monitor inbound and outbound traffic
- ✓ Detect attacks, scans, and policy violations
- ✓ Identify malicious or suspicious IP addresses
- ✓ Investigate security incidents
- ✓ Ensure compliance and network security



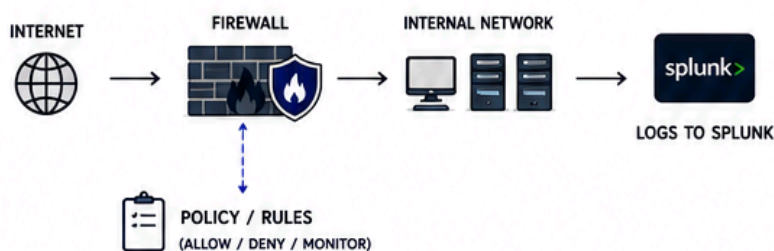
KEY TAKEAWAY

Firewall logs provide visibility into network traffic and help SOC analysts detect threats, enforce policies, and strengthen the organization's security posture.

3 PREREQUISITES

-  Splunk Enterprise installed & configured
-  Firewall Device (Palo Alto / Fortinet / Cisco)
-  Firewall logs forwarded to Splunk
-  Sourcetype configured for firewall logs

4 FIREWALL ARCHITECTURE (LOG FLOW)



5 ALLOW vs DENY

✓ ALLOW (PERMIT)

- Traffic matches an allow rule
- Connection is permitted
- Logged as allowed
- Example: HTTPS traffic to web server

✗ DENY (BLOCK)

- Traffic matches a deny rule
- Connection is blocked
- Logged as denied
- Example: RDP from unknown IP

6 SAMPLE FIREWALL LOG (PALO ALTO FORMAT)

```

Receive Time: 05/18/2024 12:45:32
Type: traffic
Subtype: end
Action: allow
Rule: allow-web
Source IP: 192.168.1.105
Destination IP: 203.0.113.25
Source Port: 52341
Destination Port: 443
Protocol: tcp
Application: web-browsing
Bytes Sent: 2456
Bytes Received: 12543
Session ID: 123456789
Device: PA-VM
Virtual System: vsys1
NAT Source IP: 192.168.1.105
NAT Destination IP: 203.0.113.25
  
```

Field Breakdown

FIELD	DESCRIPTION
Receive Time	Time the log was generated
Action	allow / deny
Rule	Firewall rule name hit
Source IP	Originating IP address
Destination IP	Target IP address
Source Port	Origin port number
Destination Port	Target port number
Protocol	TCP / UDP / ICMP
Application	Application identified
Bytes Sent/Received	Data transferred
Session ID	Unique session identifier

7 COMMON FIREWALL LOG SOURCES

- Palo Alto Networks
- Fortinet FortiGate
- Cisco ASA / Firepower
- Check Point
- SonicWall
- pfSense / OPNsense

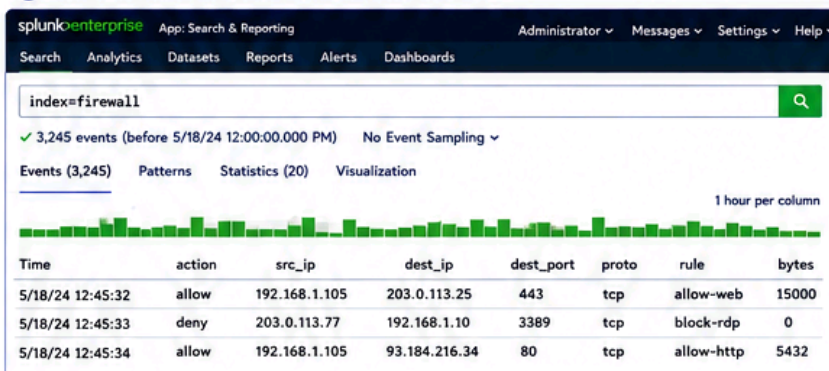


9 IMPORTANT FIREWALL LOG FIELDS

FIELD	DESCRIPTION
time	Timestamp of the event
action	Action taken (allow/deny)
rule	Rule name or ID that matched
src_ip	Source IP address
src_port	Source port number
dest_ip	Destination IP address
dest_port	Destination port number
proto	Protocol (tcp/udp/icmp)
application	Application identified
bytes	Total bytes transferred



8 VIEW IN SPLUNK (SEARCH & REPORTING APP)



10 KEY TAKEAWAY



Firewall log analysis helps SOC analysts monitor network traffic, detect malicious activity, enforce security policies, and provide critical insights during investigations.


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DHCP LOG ANALYSIS (HANDS-ON)


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LET'S ANALYZE DHCP LOGS IN SPLUNK

1 UPLOAD DHCP LOGS TO SPLUNK

- Go to **Settings** (gear icon) > **Add Data**
- Click **"Upload"**
- Drag and drop your DHCP log file or click **"Select files"**
- Set Source Type: **dhcp**
- Set Index: **dhcp**
- Click **"Next"** and review settings
- Click **"Submit"**

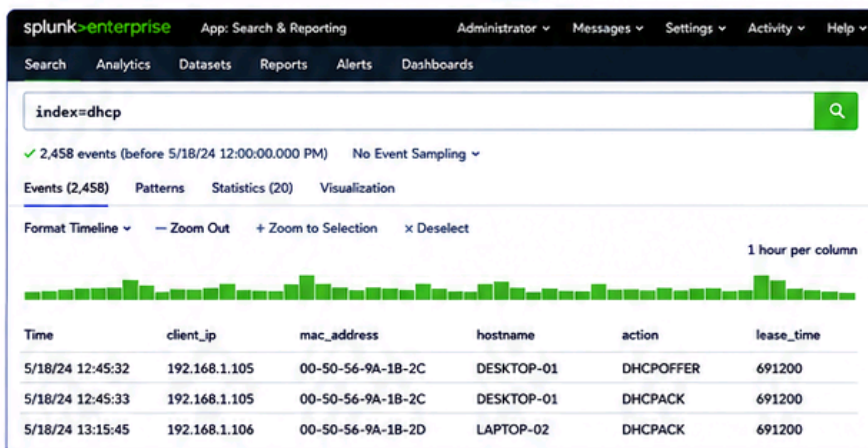

 Upload
DHCP Log File

TIP: Use sourcetype **dhcp** (or custom) for better parsing and search.

2 VERIFY DHCP LOGS

Search & Reporting app > Enter the search:

index=dhcp



3 10 USEFUL SPL QUERIES FOR DHCP LOGS

#	SPL QUERY	PURPOSE
1	index=dhcp	Show all DHCP events
2	index=dhcp stats count by action	Count by DHCP action
3	index=dhcp stats count by client_ip	Top clients by activity
4	index=dhcp action="DHCP OFFER"	See all IP offers
5	index=dhcp action="DHCP ACK"	See all IP leases
6	index=dhcp action="DHCP NAK"	Find IP assignment failures
7	index=dhcp stats dc(mac_address) by client_ip	Detect duplicate IP usage
8	index=dhcp rare client_ip	Find rare/unusual clients
9	index=dhcp timechart count by action	DHCP activity over time
10	index=dhcp stats min(lease_time) as min_lease max(lease_time) as max_lease	Lease time analysis

4 DETECT ROGUE DHCP SERVERS



Identify DHCP offers from unauthorized servers.

```
index=dhcp action="DHCP OFFER"
| stats dc(server_ip) as dhcp_servers by server_ip
| where dhcp_servers > 1
```

5 DETECT UNKNOWN DEVICES



Find clients that are not in the known device list (lookup table).

```
index=dhcp action="DHCP ACK"
| lookup known_devices.csv mac_address OUTPUT device_name
| where isnull(device_name)
```

6 DETECT IP CONFLICTS



Detect multiple MAC addresses using the same IP address.

```
index=dhcp action="DHCP ACK"
| stats dc(mac_address) as mac_count by client_ip
| where mac_count > 1
```

7 LEASE EXPIRATION MONITORING



Find leases that are about to expire.

```
index=dhcp action="DHCP ACK"
| eval expire_time = lease_time + _time
| where expire_time < now() + 3600
| table client_ip, mac_address, expire_time
```

8 TOP IP CONSUMERS



Top clients by number of leases.

```
index=dhcp action="DHCP ACK"
| stats count by client_ip, mac_address
| sort - count
| head 10
```

9 FAILED DHCP ASSIGNMENTS



Find failed or declined IP assignments.

```
index=dhcp action IN ("DHCP NAK", "DHCP DECLINE")
| stats count by client_ip, mac_address, server_ip
| sort - count
```

10 INVESTIGATION WORKFLOW


 Alert
Triggered

 Review DHCP
Events

 Identify Client
& Server

 Analyze Action
(Offer, Ack, Nak)

 Correlate with
Other Logs

 Determine
Impact

 Document &
Report

11 SOC ANALYST CHECKLIST



- ✓ Verify DHCP server IPs
- ✓ Check for rogue DHCP servers
- ✓ Review unusual client activity
- ✓ Validate IP lease duration
- ✓ Look for IP conflicts
- ✓ Correlate with firewall, DNS, and authentication logs
- ✓ Document findings and escalate if required

12 KEY TAKEAWAY



DHCP log analysis helps SOC analysts maintain network visibility, detect rogue DHCP servers, identify unknown devices, prevent IP conflicts, and ensure efficient IP address management across the organization.



FIREWALL LOG ANALYSIS (HANDS-ON)

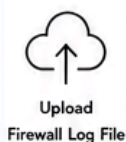

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LET'S ANALYZE FIREWALL LOGS IN SPLUNK

1 UPLOAD FIREWALL LOGS TO SPLUNK

- Go to **Settings** (gear icon) > **Add Data**
- Click **"Upload"**
- Drag and drop your firewall log file or click **"Select files"**
- Set Source Type: **firewall**
- Set Index: **firewall**
- Click **"Next"** and review settings
- Click **"Submit"**

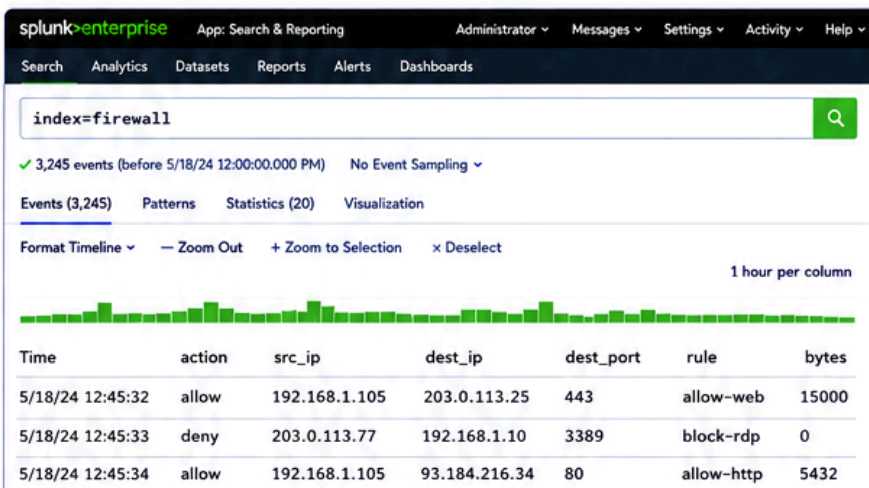


TIP: Use sourcetype **firewall** (or custom) for better parsing and search.

2 VERIFY FIREWALL LOGS

Search & Reporting app > Enter the search:

index=firewall



3 10 USEFUL SPL QUERIES FOR FIREWALL LOGS

#	SPL QUERY	PURPOSE
1	index=firewall stats count by action	Show allowed vs denied traffic
2	index=firewall action=deny table _time, src_ip, dest_ip, dest_port, rule	Show all denied events
3	index=firewall stats count by src_ip sort - count	Top source IPs
4	index=firewall stats count by dest_ip sort - count	Top destination IPs
5	index=firewall stats count by dest_port sort - count	Top destination ports
6	index=firewall dest_port=22 OR dest_port=3389	Detect SSH/RDP access attempts
7	index=firewall action=deny stats count by src_ip, dest_port	Top blocked ports by source IP
8	index=firewall timechart count by action	Traffic trend over time
9	index=firewall rare src_ip	Find rare/odd source IPs
10	index=firewall rule="allow-web" stats sum(bytes) as total_bytes by dest_ip	Bandwidth usage by destination

4 DETECT PORT SCANNING



Identify multiple connection attempts to many ports.

```
index=firewall action=deny
| stats dc(dest_port) as unique_ports, count as attempts
  by src_ip
| where unique_ports > 20
| sort - attempts
```

5 DETECT BLOCKED CONNECTIONS



Find frequently blocked connections.

```
index=firewall action=deny
| stats count by src_ip, dest_ip, dest_port
| sort - count
```

6 GEOIP ANALYSIS



Analyze traffic based on country.

```
index=firewall
| iplocation src_ip
| stats count by Country
```

7 TOP SOURCE IPS



Find the most active source IPs.

```
index=firewall
| stats count by src_ip
| sort - count
| head 10
```

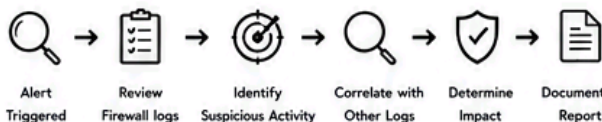
8 TOP DESTINATION IPS



Find the most contacted destination IPs.

```
index=firewall
| stats count by dest_ip
| sort - count
| head 10
```

9 INVESTIGATION WORKFLOW



10 SOC ANALYST CHECKLIST



- ✓ Verify source of the event
- ✓ Check timestamps and time zone
- ✓ Analyze source IP and reputation
- ✓ Review destination IP and port
- ✓ Validate firewall rule and action
- ✓ Correlate with proxy, DNS, and authentication logs
- ✓ Determine if traffic is legitimate or malicious
- ✓ Document findings and escalate if required

11 KEY TAKEAWAY



Firewall log analysis helps SOC analysts detect attacks, unauthorized access, policy violations, and suspicious traffic. By using the right SPL queries and investigation workflow, analysts can strengthen the organization's security posture and respond effectively to threats.



SOC ANALYST CHEAT SHEET



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QUICK REFERENCE GUIDE

Essential commands, ports, workflows, and checklists every
SOC Analyst should know!

1 ESSENTIAL SPL COMMANDS

COMMAND	DESCRIPTION
search	Search for events that match keywords or conditions
stats	Generate statistical reports
chart	Create charts and visualizations
timechart	Show trends over time
top	Show top values for a field
rare	Find rare or unusual values
dedup	Remove duplicate events
rex	Extract fields using regular expressions
eval	Create new fields or evaluate expressions
lookup	Enrich events using a lookup table
transaction	Group related events into transactions
table	Display events in a table format
sort	Sort results by a field
where	Filter events using conditions
fields	Select specific fields to display
join	Join events from subsearch
append	Combine results from multiple searches

2 COMMON PORTS (REFERENCE)

SERVICE	PORT	PROTOCOL	DESCRIPTION
DNS	53	UDP/TCP	Domain Name System
FTP	21	TCP	File Transfer Protocol
SSH	22	TCP	Secure Shell
SMTP	25	TCP	Simple Mail Transfer
HTTP	80	TCP	HyperText Transfer
HTTPS	443	TCP	HTTP Secure
DHCP	67/68	UDP	Dynamic Host Config
RDP	3389	TCP	Remote Desktop
SMB	445	TCP	Server Message Block
LDAP	389	TCP/UDP	Lightweight Directory
SNMP	161	UDP	Simple Network Mgmt.

3 INVESTIGATION WORKFLOW



4 MITRE ATT&CK FRAMEWORK (TACTICS)

Initial Access	Gain initial foothold in the environment
Execution	Execute code or malicious commands
Persistence	Maintain access over time
Privilege Escalation	Gain higher level permissions
Defense Evasion	Avoid detection by defenses
Credential Access	Steal or brute force credentials
Discovery	Reconnaissance and system discovery
Lateral Movement	Move within the network
Collection	Gather data of interest
Exfiltration	Exfiltrate data from the network
Impact	Disrupt or destroy systems/data

5 LOG ANALYSIS CHECKLIST

- ✓ Verify the log source and sourcetype
- ✓ Check the timestamp and time zone
- ✓ Identify the user or service account
- ✓ Identify source IP address
- ✓ Identify destination IP address
- ✓ Review the protocol and port
- ✓ Understand the event/message
- ✓ Correlate with other relevant logs
- ✓ Determine the impact and severity
- ✓ Take appropriate response actions
- ✓ Document findings and evidence

6 COMMON LOG SOURCES

DNS	FTP	HTTP/HTTPS	SSH
SMTP	DHCP	FIREWALL	IDS/IPS
VPN	AD / WINDOWS	ENDPOINT	CLOUD

7 USEFUL SPL SNIPPETS

index=* stats count by sourcetype	# Count by sourcetype
index=* top src_ip	# Top Source IPs
index=* top dest_ip	# Top Destination IPs
index=* stats count by user	# Top Users
index=* stats count by action	# Events by Action
index=* timechart count	# Events over time
index=* rare src_ip	# Rare Source IPs
index=* stats avg(bytes) by dest_ip	# Avg bytes by Dest IP
index=* chart count by status	# Events by Status

8 EVENT SEVERITY GUIDE

SEVERITY	LEVEL	DESCRIPTION
Critical	5	Severe impact, immediate action
High	4	High impact, urgent attention
Medium	3	Moderate impact, investigate
Low	2	Low impact, monitor
Informational	1	Informational events

9 ANALYST BEST PRACTICES

- ★ Always validate before escalating
- ★ Think like an attacker
- ★ Keep learning and stay updated
- ★ Document everything
- ★ Use threat intelligence
- ★ Automate and streamline workflows
- ★ Communicate clearly with team
- ★ Focus on improvement, not perfection



REMEMBER:

Every log tells a story.
A skilled SOC Analyst knows
how to read it.

**"EVERY LOG TELLS A STORY.
A SKILLED SOC ANALYST
KNOWS HOW TO READ IT."**



TUNNEL LOG ANALYSIS


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UNDERSTANDING TUNNEL LOGS

1 WHAT ARE TUNNEL LOGS?

Tunnel logs are generated when data is encapsulated within other protocols to create encrypted or virtual communication channels between endpoints.

2 WHY TUNNEL LOGS MATTER IN SOC

- ✓ Attackers use tunnels to bypass firewalls and IDS/IPS.
- ✓ Tunnels hide actual payload and make detection difficult.
- ✓ Can be used for data exfiltration and C2 communication.
- ✓ Helps in identifying unauthorized VPNs and rogue tunnels.



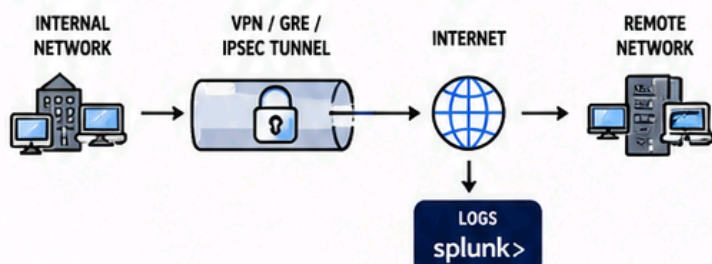
Key Takeaway

Tunnel logs help SOC analysts identify encrypted communication channels used by attackers to bypass network controls and exfiltrate data.

3 PREREQUISITES

- Splunk Enterprise installed & configured
- Zeek IDS configured and forwarding logs
- Tunnel logs forwarded to Splunk
- Sourcetype configured for tunnel logs

4 TUNNEL ARCHITECTURE (TYPICAL FLOW)



5 COMMON TUNNEL TYPES

TYPE	DESCRIPTION
GRE (Generic Routing Encapsulation)	Encapsulates network layer packets inside IP packets (Protocol 47).
IP-in-IP	Encapsulates IP packets inside another IP packet (Protocol 4).
IPsec (Internet Protocol Security)	Provides encrypted and authenticated communication using ESP/AH.
IPv6 Tunnel	Carries IPv6 traffic over IPv4 infrastructure.

6 SAMPLE TUNNEL LOG (ZEEK FORMAT)

```
2024-05-18T12:45:32.123456Z tunnel 192.168.1.10 203.0.113.5 47 GRE
PCAP 12345 67890 512 1234 0 - - - - -
```

FIELD BREAKDOWN

FIELD	DESCRIPTION
Timestamp	Time when the tunnel event was logged
uid	Unique connection ID
src_ip	Source IP address (Internal host)
dest_ip	Destination IP address (External host)
proto	Protocol used (47=GRE, 50=ESP, etc.)
service	Tunnel service (GRE, IPsec, etc.)
bytes	Total bytes transferred
packets	Total packets transferred

7 COMMON TUNNEL PORTS / PROTOCOLS

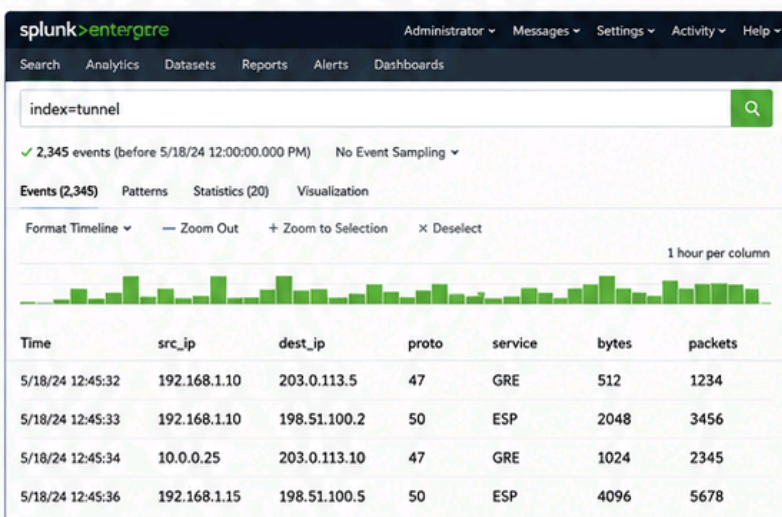
TUNNEL TYPE	PROTOCOL	IP PROTOCOL #	PORTS
GRE	GRE	47	N/A
IP-in-IP	IP	4	N/A
IPsec (ESP)	ESP	50	N/A
IPsec (AH)	AH	51	N/A
OpenVPN	UDP	17	1194
WireGuard	UDP	17	51820

9 COMMON TUNNEL LOG SOURCES

- Zeek / Bro IDS
- Palo Alto Networks Firewall
- Cisco ASA / Firepower
- Fortinet FortiGate
- OpenVPN / WireGuard
- Windows RRAS / VPN Logs
- Linux strongSwan / Libreswan (IPsec)



8 VIEW IN SPLUNK (SEARCH & REPORTING APP)



TUNNEL LOG ANALYSIS (HANDS-ON)


BLUETEAMERS

DEFEND. DETECT. RESPOND.

LET'S ANALYZE TUNNEL LOGS IN SPLUNK

1 UPLOAD TUNNEL LOGS TO SPLUNK

- Go to **Settings** (gear icon) > **Add Data**
- Click **"Upload"**
- Drag and drop your tunnel log file or click **"Select files"**
- Set **Source Type**: **tunnel**
- Set **Index**: **tunnel**
- Click **"Next"** and review settings
- Click **"Submit"**

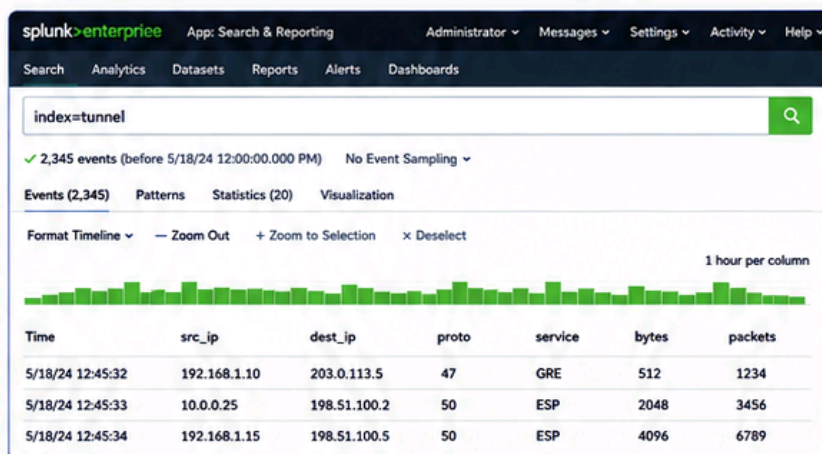


TIP: Use sourcetype **tunnel** (or custom) for better parsing and search.

2 VERIFY TUNNEL LOGS

Search & Reporting app > Enter the search:

index=tunnel



3 10 USEFUL SPL QUERIES FOR TUNNEL LOGS

#	SPL QUERY	PURPOSE
1	index=tunnel	Show all tunnel events
2	index=tunnel stats count by service	Count by tunnel service
3	index=tunnel stats sum(bytes) by src_ip	Top talkers (by bytes)
4	index=tunnel stats count by src_ip	Top source IPs
5	index=tunnel stats count by dest_ip	Top destination IPs
6	index=tunnel service=GRE	Show GRE tunnel traffic
7	index=tunnel service=ESP	Show IPsec (ESP) traffic
8	index=tunnel timechart sum(bytes)	Tunnel traffic over time
9	index=tunnel bytes>1000000	Large data transfer events
10	index=tunnel rare src_ip	Rare/Unusual source IPs

4 DETECT VPN ABUSE



Look for VPN connections from unusual locations or outside business hours.

```
index=tunnel service IN ("IPsec", "OpenVPN")
| stats count by src_ip
| sort - count
| head 10
```

5 DETECT DNS TUNNELING (POSSIBLE)



Identify unusual DNS traffic over tunnel (high queries).

```
index=tunnel service=DNS
| stats count by src_ip
| where count>1000
| sort - count
```

6 DETECT LARGE DATA TRANSFER



Detect potential data exfiltration over tunnels.

```
index=tunnel
| stats sum(bytes) as total_bytes by src_ip dest_ip
| where total_bytes>1000000
| sort - total_bytes
```

7 DETECT UNKNOWN TUNNEL ENDPOINTS



Identify new/unknown endpoints.

```
index=tunnel
| stats count by src_ip dest_ip
| lookup known_tunnel_endpoints.csv dest_ip OUTPUT is_known
| where is_known="false"
```

8 INVESTIGATION WORKFLOW



9 SOC ANALYST CHECKLIST

- ☒ Unusual tunnel type or protocol
- ☒ Tunnel from unknown/rare country
- ☒ High volume of tunnel traffic
- ☒ Large data transfer events
- ☒ New or unknown tunnel endpoints
- ☒ Correlate with firewall, proxy & DNS logs

10 KEY TAKEAWAY



Tunnel logs help SOC analysts detect VPN abuse, data exfiltration, DNS tunneling, and unauthorized communication channels. Always baseline normal traffic and alert on deviations.

SMTP LOG ANALYSIS



BLUETEAMERS

— DEFEND. DETECT. RESPOND. —

UNDERSTANDING SMTP LOGS

SMTP logs help SOC analysts monitor email traffic, detect phishing, spam, malware, and policy violations.

1 WHAT ARE SMTP LOGS?

SMTP (Simple Mail Transfer Protocol) logs record email sending, receiving, authentication, delivery status, failures, and security events on mail servers.

2 WHY SMTP LOGS MATTER

- ✓ Detect phishing and spam campaigns
- ✓ Identify malicious attachments and links
- ✓ Monitor email delivery failures and abuse
- ✓ Track unauthorized relay attempts
- ✓ Help in incident investigation & compliance



Key Takeaway

SMTP logs are critical for protecting your organization from email-based threats and ensuring secure mail communication.

3 PREREQUISITES



Splunk Enterprise installed & configured



Mail Server (Exchange / Postfix / Sendmail)

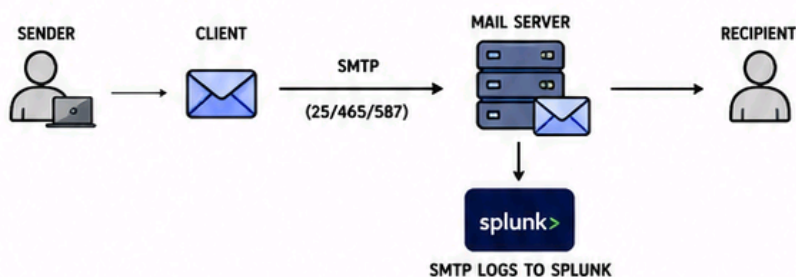


SMTP logs forwarded to Splunk



Sourcetype configured for SMTP logs

4 MAIL FLOW ARCHITECTURE



5 SMTP PORTS

PORT	DESCRIPTION
25	SMTP (Unencrypted)
465	SMTPS (SSL Encrypted)
587	SMTP Submission (STARTTLS)
2525	Alternative SMTP

6 SAMPLE SMTP LOG (POSTFIX FORMAT)

```

May 18 12:45:32 mailserver postfix/smtpd[12345]:
connect from unknown[192.168.1.55]
May 18 12:45:33 mailserver postfix/smtpd[12345]:
A1B2C3D4E5: client=192.168.1.55
May 18 12:45:34 mailserver postfix/cleanup[12350]:
A1B2C3D4E5: message-id=<20240518124534.A1B2C3@mail.com>
May 18 12:45:35 mailserver postfix/qmgr[12360]:
A1B2C3D4E5: from=<user@example.com>,
size=2345, nrcpt=1 (queue active)
May 18 12:45:36 mailserver postfix/smtpd[12345]:
A1B2C3D4E5: to=<victim@company.com>,
relay=mail.company.com[10.0.0.10]:25, status=sent (250 2.0.0 Ok)
  
```

7 COMMON SMTP LOG SOURCES

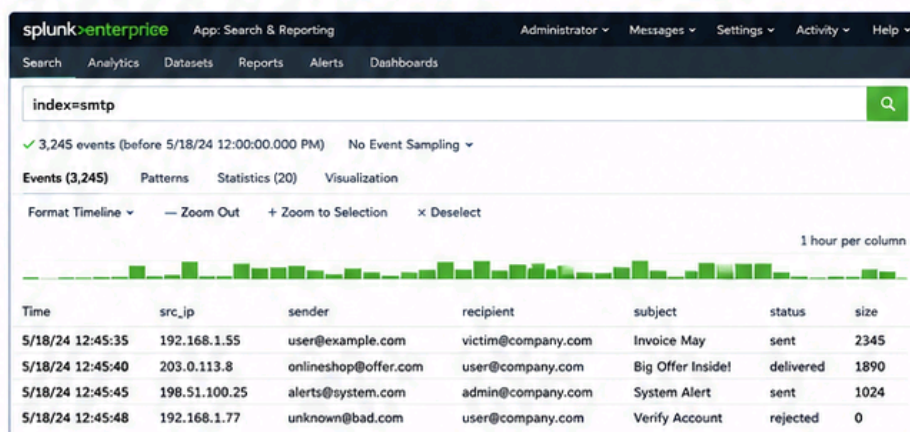
- Postfix / Sendmail
- Microsoft Exchange
- Google Workspace (Gmail logs)
- Proofpoint / Mimecast
- Barracuda Email Security Gateway
- Cisco ESA
- Office 365 / Exchange Online



8 IMPORTANT SMTP LOG FIELDS

FIELD	DESCRIPTION
timestamp	Time of the SMTP event
src_ip	IP address of the sender
src_port	Source port used
dest_ip	Mail server IP address
sender	From email address
recipient	To email address
subject	Email subject
message_id	Unique message ID
status	Delivery status (sent, failed, rejected)
size	Email size in bytes
action	Action taken (accept, reject, block, etc.)
reason	Reason for failure or rejection

9 VIEW IN SPLUNK (SEARCH & REPORTING APP)



10 COMMON SMTP THREATS DETECTED



11 KEY TAKEAWAY



SMTP log analysis helps SOC analysts detect email-borne threats like phishing, spam, and malware, and ensures secure, reliable, and compliant email communication.